Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_12

1. Summary of Key Deviations

The analysis of Run_12 reveals a critical fault localized to a single engine feed line, with the rest of the system reacting in a physically consistent manner.

- **Flow_Engine1 (Critical):** This is the primary and most severe anomaly. The signal for engine 1 shows a **major negative drift**, starting at -60.5% and culminating in a complete **signal dropout (-99.7%)** by state s5. This indicates a near-total loss of flow to one engine.
- **Flow_Bypass (Moderate):** A significant positive **offset** (up to **+20.9%**) is observed throughout the run. This is the direct result of the blockage at engine 1; the flow that cannot pass through the blocked nozzle is correctly rerouted through the bypass loop.
- **Flow_Pump (Minor):** A minor positive **offset** of approximately **+9%** is present. This is the pump working slightly harder to compensate for the downstream blockage and maintain overall system pressure.
- **Flow_Engine2, 3, 4 (Minor/Nominal):** In contrast to engine 1, the other three engines show only a minor **negative offset** (-3%) and only during the high-throttle phase (s5). For the most part, they are healthy.
- **Pressures and Flow_PRV (Nominal):** All system pressures show only minor, negligible deviations, and the PRV flow is healthy. This confirms the system is managing the fault without a dangerous over-pressurization.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Injector Nozzle 1**	**Severe Blockage / Clogging**	This is the most direct and compelling explanation. A clogged nozzle at Engine 1 would cause its specific flow reading (`Flow_Engine1`) to drop to zero. The rest of the system's reaction—a large increase in bypass flow (`Flow_Bypass`) and a slight increase in pump work (`Flow_Pump`)—is a textbook physical response to such a localized downstream blockage.

2	**Fuel Metering Unit (FMU)**	**Internal Blockage (Channel to Engine 1)**	The FMU is the component that splits the main fuel line into four separate feeds for the engines. A blockage or failure of the specific internal channel feeding Engine 1 would produce symptoms identical to a clogged nozzle. It is an equally strong candidate for the location of the blockage.
3	**Piping to Engine 1**	**Line Crimp / Blockage**	The physical fuel line running from the FMU to injector nozzle 1 could be crimped or blocked. This is physically equivalent to a clogged nozzle or FMU channel and is another possible location for the same root fault.
4	**Flow_Engine1 Sensor**	**Sensor Dropout / Failure**	A sensor-only failure is a possibility. However, it is much less likely because the other system signals (**increased bypass flow** and **increased pump flow**) are perfectly consistent with a *real* physical blockage. If the sensor had simply failed, there would be no physical reason for the rest of the system to react so strongly.
5	**System Contamination**	**Debris causing blockage**	This is the likely underlying *cause* of the blockage itself (a piece of debris lodging somewhere). The primary diagnostic task is to identify the *location* of the blockage, which is somewhere in the Engine 1 fuel path.

3. Most Probable Hypothesis

The most probable root cause is a **Severe Blockage in the fuel path to Engine 1**, located either at the **Injector Nozzle 1** or within the corresponding **FMU channel**.

Reasoning: The evidence for a severe, localized blockage is definitive. The complete dropout of `Flow_Engine1` while `Flow_Engine2`, `Flow_Engine3`, and `Flow_Engine4` remain almost entirely healthy pinpoints the failure to the individual fuel line for Engine 1. The system's response is perfectly logical: the controller increases pump flow slightly to compensate, and the un-delivered fuel is rerouted via the bypass valve, causing the `Flow_Bypass` to increase. This harmonious interplay between the other signals provides strong corroborating evidence that the `Flow_Engine1` dropout is a real physical event, not just a sensor error.